

Ecosim biomass forcing time series in Ecospace.

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Ecosim biomass forcing time series inputs can be used to force the biomass of Ecospace. When this feature is enabled the spatially averaged biomass predicted by Ecospace will be equal to the Ecosim forced biomass from the forcing time series at the current time step. The spatial distribution of biomass will still be predicted by the Ecospace dispersal model.

How it works

During the initialization of Ecospace the Capacity Model sets the capacity of each cell based on either the Foraging Response or the Habitat Base Foraging settings. Then the Ecopath biomass is spatially distributed across the map using the cell capacity and or the migration preference of each cell as a weighting factor. At the end of the initialization the spatially average biomass of Ecospace will equal the baseline Ecopath biomass.

When the Ecosim biomass forcing is enabled the biomass from the time series will be distributed across the map using the current biomass distribution pattern. This maintains the spatial distributions predicted by Ecospace but overrides the biomass in a given cell.

This takes the form

$$B_{ijk} = Bf_{kt} \frac{B_{ijk}}{\bar{B}_k}$$

where B_{ijk} is the biomass in row (i) column (j) for group (k), Bf_{kt} is the time series forced biomass for group (k) at time step (t), $\bar{B}_k$ is the mean Ecospace biomass of group (k).

How to use it

Load time series data containing biomass forcing into Ecopath via the Ecosim **Time series** interface. Only time series data define as **Biomass forcing (-1)** will be used to force the Ecospace biomass. This makes the forcing data available to Ecospace but by default it will not be enabled. To enable it check the **Use Ecosim biomass forcing** check box on the **Ecospace parameters** interface, under the **Time Series** heading. If no forcing data is loaded this check box will be grayed out. You can turn individual forcing time series on or off via the Ecosim **Time series** interface.